



FAST National University of Computer and Emerging Sciences

Thesis Proposal

Towards Improving Deep Embedded Document Clustering with long range text semantics and neighborhood documents information

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1. Abstract

Document clustering is an unsupervised machine learning problem, which aims to learn latent semantic patterns from the text data. Deep learning based document clustering algorithms recently performed very well. These algorithms try to learn the features and clustering arrangement simultaneously. The main bottleneck in these algorithms is to maintain a good balance between the representation and clustering. This research work aims to address this bottleneck with twofold objectives (i) learning long-range semantic dependencies using a Transformer-based architecture to improve document representation, and (ii) utilizing the neighborhood information in different documents to properly guide the clustering process for better structural information, for this the thesis intends to use Graph Convolutional Network (GCN) as a global information about the clustering. The whole architecture is trained with an end-to-end objective based on balancing clustering refinement, structural consistency, and contrastive alignment. The thesis intends to use the existing document clustering datasets with evaluation metrics to evaluate clustering performances, robustness and stability of clusters outcomes.

2. Introduction

Document clustering is an unsupervised learning task that aims to organize unlabeled documents into semantically coherent clusters. It plays an important role in information retrieval, topic extraction and recommendation systems and large-scale content management. However, it is difficult to cluster textual data due to the high dimensionality of document representations and the difficulty to capture meaningful semantic relationships among documents.

Recent research in deep embedded clustering (DEC) frameworks have improved clustering performance by learning feature representations and clustering assignments simultaneously. Models such as SDCN [1], DSEDC [2], and AGCN [3] proved that integrating structural relationships with graph convolutional networks (GCNs) improves clustering quality. Likewise, transformer-based approaches such as SDEC [4] and DTSN [5] enhanced semantic representation by capturing long-range contextual dependencies between documents. Despite these advancements, there are two major challenges.

First, many graph-based clustering models heavily rely on fixed graph structure constructed early in training. These static structural assumptions may introduce early-stage noise and fail to capture as semantic embeddings evolve during optimization. Second, although semantic and structural representations are often optimized simultaneously, explicit mechanisms to enforce consistency between these two views are rare.

These limitations reveal a fundamental bottleneck in deep embedded document clustering, namely maintaining an effective balance between semantic representation learning and structurally guided clustering alignment while ensuring robustness.

To address these limitations, the thesis proposes two major ideas:

- i. Obtain the long-range semantic representations using a Transformer-based framework, thus improving the representation of the content of documents.
- ii. Use neighborhood structural information via adaptive graphs construction and Graph Convolutional Network modelling to guide the refinement of clustering while training.

3. Literature Review

Initial advances in document clustering were concerned with the use of the improved text representations. The article "*The performance of BERT as data representation of text clustering*" [9] demonstrated that contextual BERT embeddings had remarkable results in comparison with other feature representations like TF-IDF on various clustering methods. It also pointed out that performance of clustering module is very sensitive to pooling and normalization strategies. This shows that the quality of representation does not ensure stable clustering. In order to mitigate this situation, the frameworks of deep embedded clustering (DEC) were proposed. Structural Deep Clustering Network (SDCN) [1] offered a joint optimization model that combines the autoencoders-based representation learning with clustering by using a student's t-distribution based soft assignments and KL divergence. This represented a shift from two stage clustering to a unified end to end optimization technique. This was later further extended in subsequent work like Deep Structural Enhanced Network for Document Clustering (DSEDC) [2], which added further structural information in the form of graphs to the learning process and enhanced the strength to cluster. Improved Deep Embedded Clustering with Local Structure Preservation (IDEC) [11] prolonged the original DEC idea by jointly optimizing the clustering objective and reconstruction loss of an under-complete autoencoder to preserve local manifold structure. IDEC, in contrast to DEC, integrates a KL-divergence clustering loss with reconstruction constraints, results in improving clustering accuracy and stability across different datasets such as; MNIST, USPS, and REUTERS.

The establishment of structural relations between documents became one of the leading strategies in enhancing the quality of clustering. Graph Convolutional Networks (GCNs) were combined with autoencoders in SDCN [1] to learn deep structural information. Similarly, DSEDC [2] used the semantic representations and structural features learned by GCNs with external, internal semantics, and optimized them together. Attention based Graph Clustering Network (AGCN) [3] added attention to dynamically combine attribute and structural representations, which overcomes over-smoothing in graph learning. These studies indicate that the use of neighborhood information can improve the performance of clustering. In the paper Feature-aware Unsupervised Learning with Joint Variational Attention and Automatic Clustering [13], a variational attention mechanism with joint optimization framework and automatic clustering were proposed. It trained latent feature distributions and attention vectors together with a KL-based clustering constraint, therefore showing better ACC and NMI compared to DEC, IDEC, and VaDE but with multi-loss

objectives. They however also report that they are sensitive to the quality of the constructed graphs and that noise propagation may occur along structural layers. More recently, Graph-Convolutional Networks: Named Entity Recognition and Large Language Model Embedding in Document Clustering [6] gave embedding based on the LLM and entity-based graph building, with better results compared with conventional KNN-based techniques. However, this method involves delicate threshold tuning when constructing graphs, which suggests that structural modeling is hyperparameter sensitive. ARGAn and ARVGA frameworks [14], proposed adversarial regularized graph autoencoders with graph convolutional network encoders. These models use adversarial training methods to impose consistency between latent embeddings and known prior distributions. Empirical findings show that both robustness and clustering performance are improved on citation network data. The method, however, has a high computing complexity during training as well as it is sensitive to the architecture of the decoder and the selection of the prior distribution.

Transformer-based embeddings have become central to document clustering with the success of contextual models as they have effectiveness beyond clustering tasks. A BERT-based framework with attention aggregation and GRU-derived patient embeddings [16] achieved high precision and F1-scores in supervised medical report classification by reinforcing the capacity of contextual transformers to capture long-range semantic dependencies besides being limited to domain-specific supervised settings. BERT embeddings were combined with a DEC-style clustering system with Semantic Deep Embedded Clustering (SDEC) [4], which optimized representation learning along with cluster assignments. This was better at semantic preservation, and was said to be expensive to compute and sensitive to initialization. Further Structural Semantic Network Deep Transformer-based Structural Semantic Network (DTSN) [5] added GCN-based structural modeling to Transformer encoders and combined semantic and structural embeddings in a combined optimization framework. Although it has shown good clustering performance, it has been reported to have weaknesses such as relying on the quality of KNN graph, scaling limitations and assumption of static embedding. A recent hybrid RNN-VAE framework [15] combined sequential dependency modeling using recurrent neural networks and latent feature learning using variational autoencoders for clustering documents. The hybrid architecture worked very well for certain datasets like BBC Sports and Reuters; the ARI and accuracy was high. It however had issues in dealing with highly heterogeneous corpora such as 20-Newsgroups and this indicates that it cannot generalize well with varying semantic distributions. These observations highlight the need for dynamically reconstructed neighborhood graphs that evolve alongside semantic representations during training.

In addition to direct graph modeling, other works enhanced clustering through better embedding geometry. These were complemented with Deep Multi-View Document Clustering with Enhanced Semantic Embedding (MDCE) [7], which enhanced semantics by relying on neighbor-based and cross-view autoencoders, then clustering with the DEC approach. This method solved the view inconsistency and resulted in the enhancement of convergence stability. Neighborhood Preserving Embedding (NPE) was introduced by the

refined SBERT [8] to re-embed sentence representations into a manifold space, followed by the local geometric structure, and then subjecting the inputs to K-means clustering. Though effective to enhance semantic consistency, it needs extra computational procedures and hyperparameter optimization. A generative clustering approach proposed by Variational Deep Embedding (VaDE) [12], integrates a Variational Autoencoder with a Gaussian Mixture prior, modeling the data generation process explicitly and performing clustering through ELBO maximization. This approach outperformed DEC and traditional baselines across multiple datasets and introduced cluster conditional sample generation capability while requiring careful pretraining to avoid poor local minima. The approaches demonstrate that the performance of clustering can be improved by the improved structure and geometry even in the absence of explicit structural graphs.

Recently, contrastive learning has been used to develop better representations. Domain-Adapted Contrastive Learning [10] used contrastive goals together with domain adaptation to enhance low-resource language clustering. Though it greatly enhanced the embedding quality, clustering was carried out in a sequential manner through HAC and K-means and not in an integrated deep clustering approach. This indicates that contrastive learning enhances semantic representation but it is seldom directly incorporated into joint semantic-structural deep clustering systems.

There are a number of recurring trends in the literature:

- ❖ Student’s t-distribution and KL divergence to cluster into deterministic coherent clusters are popularly used in the DEC-style [1], [2], [3], [4], [5], [7].
- ❖ GCNs improve the quality of clustering, yet they are vulnerable to the quality of graph construction [1], [5], [6].
- ❖ Transformer-based embeddings are better at modeling semantics over long distances, but they present computational complexity and stability issues [4], [5].
- ❖ Contrastive learning yields better representation matching but is not used with unified deep clustering optimization [10].
- ❖ Multi-view and manifold refinements are better at embedding geometry, but nonetheless do not resolve dynamic structural learning [7], [8].

These results suggest that, although semantic modeling and structural learning have been considerably developed, current options tend to use the neighborhood graph as a static element and insufficient interaction of both semantic and structural perspectives. This encourages more studies on the possibility of coming up with unified models that dynamically model document relationships and still be semantically consistent.

4. Literature Summary Table

Reference	Year	Merits	Demerits	Dataset	Evaluation Metrics
[1]	2020	Integrates AE and GCN in unified deep clustering, dual self-supervised KL optimization and strong gains across 6 datasets.	Sensitive to KNN graph quality, added model complexity and hyperparameter tuning.	USPS HHAR Reuters ACM DBLP Citeseer	ACC NMI ARI F1
[2]	2023	Enhances AE semantics using GCN structural info, layer-wise ensemble reinforcement and strong multi dataset improvement.	Early/late GCN layers may introduce noise and performance degradation on large/dense graphs.	ACM Citeseer Reuters BBC Aminer-s	ACC NMI ARI
[3]	2021	Attention based fusion of attribute and structural features, multiscale representation learning and robust on low quality graphs.	Limited gains on well-connected graphs and middle layers prone to over-smoothing.	USPS HHAR Reuters ACM Citeseer DBLP	ACC NMI ARI F1
[4]	2025	Joint transformer based semantic learning and DEC clustering, semantic refinement improves performance and SOTA on 5 benchmarks.	Requires predefined cluster number, high computational cost and sensitive to weights initialization.	AG News Yahoo! Answers DBPedia Reuters-2 Reuters-5	ACC NMI ARI
[5]	2025	Transformer GCN fusion for inter-document semantics, layer-wise fusion with reinforcement learning and strong results on BC/ACM.	Dependent on KNN graph quality, scalability/generalization issues and static embeddings.	BBC ACM	ACC NMI ARI

[6]	2025	NER based graph construction with LLM embeddings, joint embedding and clustering optimization and outperforms KNN graphs.	Graph sparsity sensitive to thresholding requires careful tuning of τ and p .	BBC News MLSUM Arxiv-10 PubMed	ACC NMI ARI
[7]	2021	Multi-view semantic enhancement via NAE/CAE, DEC-based joint optimization and stable convergence.	Higher computational cost, weaker on small/high-class datasets.	Aminer Aminer (700) 3-sources Reuters Movies multi-source news	NMI ACC
[8]	2023	Manifold refinement of SBERT embeddings, improved semantic consistency and consistent gains across tasks.	Hyperparameter sensitive and added computational cost.	STS12-16 STS-B SICK-R MR CR SUBJ MPQA SST2 TREC MRPC 20 Newsgroups	Spearman ACC NMI ARI
[9]	2022	Systematic BERT vs TF-IDF comparison, strong gains in 28/36 metrics and highlights pooling/normalization impact.	Inconsistent gains on R2 with DEC/IDEC, sensitive to normalization strategy.	AG News (4000) Yahoo! Answers (10000) Reuters R2 (5859)	ACC NMI ARI
[10]	2025	Domain adapted contrastive learning, large scale Greek corpora and approx. 50% clustering gains.	Heavy domain dependence, high computational cost, hyperparameter sensitive and not integrated deep clustering.	Greek internet/social/press corpora (1.59M 230,530 triplets 283,052 docs)	SC Cosine Manhattan Euclidean

[11]	2017	Preserves local structure via reconstruction, joint KL and AE loss and outperforms DEC.	Slower convergence requires γ tuning.	MNIST (70000) USPS (9298) REUTERS-10K (10000)	ACC NMI
[12]	2017	VAE with GMM prior for generative clustering, strong ACC across 5 datasets and cluster conditional generation.	Requires pretraining, risk of poor local minima in early training.	MNIST HHAR REUTERS-10K REUTERS STL-10	ACC KNN err
[13]	2020	Joint variational attention and clustering, multi-loss optimization and superior ACC/NMI over DEC/IDEC/VaDE.	Hyperparameter λ sensitive and complex multi-loss balancing.	Fashion-MNIST CIFAR-10 USPS 20NEWS REUTERS Stack Overflow	ACC NMI
[14]	2019	Adversarial regularization improves embedding robustness with superior link prediction and clustering.	Increased training time, sensitive to decoder/prior choice.	Cora Citeseer PubMed	ACC NMI F1 ARI AUC AP
[15]	2024	Hybrid RNN-VAE captures sequential semantics, strong ARI/ACC on Sports/Reuters and robust to noise.	Poor generalization on heterogeneous datasets (20-Newsgroups).	BBC Sports Reuters-21578 20-Newsgroups	ARI NMI Completeness Homogeneity V-measure ACC
[16]	2023	Effective contextual modeling with BERT+GRU, high P/R/F1 in medical classification.	Data specific, requires larger validation and not clustering.	1000 cancer radiology reports	ACC Precision Recall F1

5. Research Gap and Research Questions

Maintaining an effective balance between representational learning and clustering refinement remains an unresolved problem in deep embedded clustering. As improvements in either side of the problem do not necessarily result in corresponding improvements in the stability of the final clusters. Although there has been a lot of advancement in deep document clustering, there are still a number of methodological limitations that have not been addressed.

First, most of the deep embedded clustering models are based on the graph constructions that are constructed statically to model structural relationships among documents. The models that combine Graph Convolutional Networks (GCNs) with semantic encoders include SDCN [1], DSEDC [2], and Deep Transformer-based Structural Semantic Network [5], all of which have been found to be sensitive to the quality of the KNN graphs and structural noise. As these graphs are built prior to or early in training, they are not necessarily indicative of improved semantic representations that are learned during optimization. This design is rigid and constrained and can potentially spread noise at an initial level to all of training.

Second, despite being able to preserve rich long-range semantic dependencies, Transformer-based models like SDEC [4] and DTSN [5] are designed to enhance the quality of their representation, but do not fully resolve the issue related to the evolving relationship between semantic embeddings and neighborhood structure. Semantic learning and structural modeling are typically combined, or co-optimized, but explicit mechanisms to enforce semantic-structural consistency via contrastive alignment from DEC's perspective is still rare.

Third, the issue of contrastive learning has been addressed to improve the alignment of the representations, as demonstrated in Domain-Adapted Contrastive Learning [10]. Nevertheless, these methods use conflicting goals sequentially, and place them not in an integrated deep embedded clustering system that optimizes semantic modeling, structural learning, and clustering refinement simultaneously.

Also, the current graph-based clustering algorithms are also vulnerable to the hyperparameter and threshold choice when building the graph as documented in [1], [5], and [6]. This indicates limitations in existing neighborhood modeling strategies.

In general, existing methods (1) use fixed structural graphs, (2) do not provide explicit semantic-structural alignment schemes or (3) incorporate contrastive learning outside of the optimization of the clustering process. Limited research has explored frameworks that dynamically update the structures of neighborhoods during training while establishing semantic-structural consistency in a deep embedded clustering task.

In order to fill the above research gaps, the following research questions can be suggested.

1. Do periodic reconstructions of the K-nearest neighbor (KNN) graph during training result in a better performance in clustering than using the fixed graph construction in deep embedded document clustering models?
2. Does adding cross-view contrastive learning between Transformer-based semantic embeddings and GCN-based structural embeddings improve cluster separability and consistency of representations?
3. Does a dynamic neighborhood model enhance robustness to graph noise and sensitivity to hyperparameter choice in document clustering tasks?

To address these research questions, the proposed framework will implement periodic reconstruction of K-nearest neighbor (KNN) graph during the training process, cross-view contrastive alignment between semantic and structural embeddings, and the model optimization based on a unified clustering goal. The effectiveness of each of the component will be tested with strictly controlled ablation comparing both static and dynamic graph construction, with and without contrastive learning. Clustering performance will be measured via standard metrics such as Clustering Accuracy (ACC), Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI).

6. Problem Statement

Existing DEC algorithms enhance learning of representations via structural graph modeling and semantic embedding. Nonetheless, numerous methods are based on fixed neighborhood graphs, and do not explicitly align semantic and structural representations, which are sensitive to graph noise, and constrain flexibility during training. It is necessary to have a common clustering model that dynamically redefines the relationships of documents and imposes semantic-structural consistency to enhance robustness and stability of clustering.

7. Proposed Research Methodology

1. **Benchmark Dataset:** Evaluation and comparison will be conducted using datasets in public document clustering (e.g., BBC, Reuters, ACM, AG News).
2. **Data Preparation:** Raw documents will be cleaned, and tokenized by Transformer tokenizer before being converted into model-ready inputs.
3. **Semantic Representation Learning:** Transformer encoder will generate contextualized document embeddings representing long-range semantic dependencies that will allow representation learning to adapt to the joint optimization objective that combines clustering and structural objectives.
4. **Dynamic KNN Graph Construction:** A KNN graph based on similarity will be built using semantic embeddings and periodically reconstructed every k epoch(s) during training.
5. **Structural Representation Learning through GCN:** The GCN will learn the global structural relationships among documents by propagating information across dynamically reconstructed graphs.
6. **Cross-View Contrastive Alignment:** A batch contrastive loss will be used to match semantic and structural embeddings in order to impose representation consistency.
7. **Representation Fusion:** The semantic and structural representations will be fused to get unified document representations.
8. **Deep Embedded Clustering:** The Student's t-distribution-based soft assignment with the optimization of the KL-divergence will optimize the cluster assignment.
9. **Joint Optimization:** The model will be trained in an end-to-end fashion with a composite clustering, structural and contrastive loss.
10. **Model Evaluation:** The ACC, NMI and ARI will be used to evaluate performance of clustering, while robustness and stability will be assessed through multiple random seed experiments and sensitivity analysis to graph construction parameters.

8. Block diagram of proposed research methodology

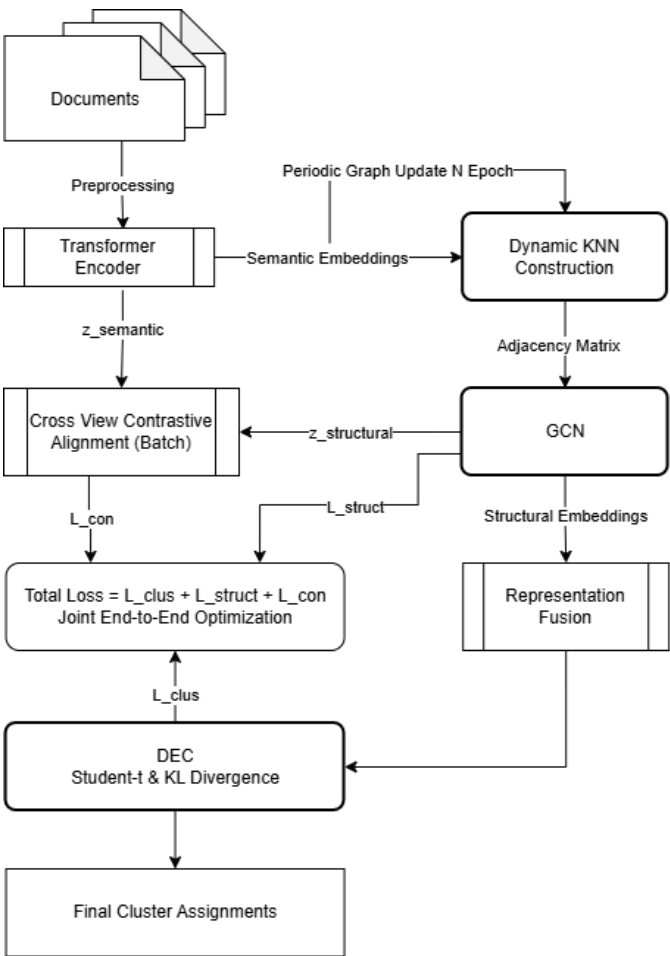


Figure 1: Flow Diagram of proposed method

9. Project Timeline

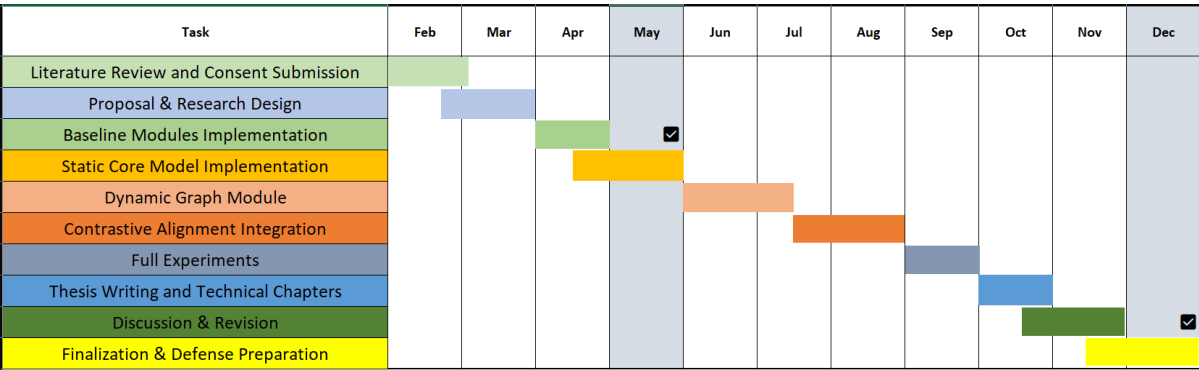


Figure 2: Project Tentative Timeline Gantt

10. References

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